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Performance monitoring of improvisation and score-playing in a turn-taking piano duet: An EEG study using altered auditory feedback

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Abstract

In music ensemble performance, perception–action coupling enables the processing of auditory feedback from oneself and other players. However, improvised actions may affect this coupling differently from predetermined actions. This study used two-person EEG to examine how pianists responded to altered pitch feedback to their own or their partner's actions while they alternated scores or improvised melodies. Feedback-related negativity (FRN) response for self-action was greater in scored than improvised conditions, indicating the enhanced action encoding by playing the score. However, subsequent P3a and P3b responses for self-action were not different across score and improvisation. Further, the P3b response was greater when the two pianists exchanged similar types of melodies (i.e., both improvised or both scores) compared with different types of melodies, suggesting that later cognitive processes may be associated with the task relevance or level of jointness. The presence of the FRN and P3 complex in self-generated improvised action points to the dynamic nature of performance monitoring even without preconceived action plans. In contrast, the FRN and P3 complex in partner-generated improvised actions were subdued compared to the baseline, likely due to the unpredictable nature of the improvised actions of others. Finally, we found a tendency that higher trait empathy was associated with smaller self-action FRN, possibly implying musicians' prioritization of joint goals. Overall, our results suggest that improvisation in a musical turn-taking task may be distinct from score-playing for the earlier processes of performance monitoring, whereas later processes might involve updating a joint representation of the musical context.

KEYWORDS

electroencephalography (EEG), event-related potentials (ERP), improvisation, joint actions, performance monitoring, empathy

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1 | INTRODUCTION

Humans interact with each other daily for many different tasks. When having a conversation, passing a basketball, or playing instruments together, we coordinate by monitoring our own intentions and actions as well as those of others. Collaborating in complex activities with shared goals and intentions is an important capability of the human species (Tomasello et al., 2005). To support this capability, our brain must represent others' goals to predict action outcomes (usually sensory feedback information), in a joint task (Sebanz et al., 2006). However, unlike perfectly choreographed or routine actions, most daily collaborative activities require us to make decisions on the fly. Whether it be choosing which words to say, which angle to throw a ball from, or which notes to play next, we improvise, continuously responding and contributing to the given social or environmental context. During those interactions, sequences of perception, prediction, and improvised action occur within and between participating agents. Such processes during music improvisation are of particular interest because spontaneous decisions have to be made during musical performance not only from social- and goal-oriented perspectives but also from creative and aesthetic perspectives. Previous studies have shown stronger neural signatures of harmonic expectation and auditory sequence processing in improvisation experts (Bengtsson et al., 2007; Berkowitz & Ansari, 2008; Donnay et al., 2014; Goldman et al., 2020; Limb & Braun, 2008; Lopata et al., 2017; Przysinda et al., 2017; see Beaty, 2015 for review). However, only a few studies have examined the neural correlates of musical action and perception during improvised performance. When improvising, musicians must dynamically manage various processes, such as prediction, planning, action, and perception, individually and incorporate information from fellow musicians as the improvisation is part of a joint musical pursuit.

To date, only a small number of studies have examined joint improvisation with respect to its behavioral, physiological, and neural correlates. In jazz piano duets, successful musical improvisation features increased temporal alignment and harmonic consonance between players (Setzler & Goldstone, 2020). Enhanced between-partner temporal synchrony and a sense of togetherness are further correlated with increased heart rates and motion intensities (Noy et al., 2015). Joint improvisation compared to solo playing also induces intrabrain connections as indexed by greater beta power (around 15 Hz) (Müller et al., 2013). Joint improvisation can be found in nonmusical domains such as language as well. During a joint spontaneous sentence production task, event-related potential (ERP) components like N400 and P600 have been found in response to unexpected words (Fjaellingsdal et al., 2020). Despite the sparseness of studies about joint improvisation in the literature, extant work points to the importance of task-specific demands, the prediction and perception of action outcomes, and the roles of the co-actors in a shared, goal-oriented task.

Our particular interest lies in the process of *performance monitoring*, whereby the brain predicts one's action outcome and integrates its resultant sensory feedback through the *perception-action cycle* (Holroyd & Coles, 2002; Loui et al., 2008; Prinz, 1997; see Ridderinkhof et al., 2004 for review), which encompasses psychological operations such as performing an action, perceiving feedback, evaluating its success or failure, and refining the subsequent predictions and actions. The electrophysiological correlates of performance monitoring can be investigated with ERP measurements. Specifically, the feedback-related negativity (FRN), which occurs in the frontocentral area of the brain around 200 ms after the feedback onset, is typically associated with how the action outcome deviates from the intended outcome expected by the agent who performed that action (Bellebaum & Colosio, 2014; Gehring & Willoughby, 2002; Holroyd & Coles, 2002; Huberth et al., 2019; Katahira et al., 2008; Loehr et al., 2013; Maidhof et al., 2010; Mathias et al., 2019). The FRN is often followed by a P3 complex consisting of a frontocentral P3a and parietal P3b. P3a relates to attentional mechanisms on the stimuli in task processing, while P3b relates to attention and subsequent memory processing (see Polich, 2007 for review). The P3 amplitude becomes higher when the stimuli are more unexpected (Hajcak et al., 2005, 2007). Furthermore, P3b is sensitive to the difference between joint and individual tasks (Demiral et al., 2016; Kato et al., 2016; Kourtis et al., 2013; Peterburs et al., 2019). This performance monitoring process is crucial for fulfilling a joint goal in accordance with reinforcement learning theory (see Ullsperger et al., 2014 for review).

Altered auditory feedback during piano performance has consistently elicited the FRN/P3 complex in the literature (Huberth et al., 2019; Katahira et al., 2008; Loehr et al., 2013; Maidhof et al., 2010; Mathias et al., 2019). In solo piano performances, auditory feedback that deviated from the intended action outcome elicited FRN, when the note in feedback was altered by a semitone (Katahira et al., 2008; Maidhof et al., 2010) or to another diatonic scale note (Mathias et al., 2019). Mathias et al. (2019) provided altered feedback with notes that were one or two ahead in a piano sequence and found similar FRN amplitudes across positions, suggesting that FRN is more salient to performers' action-related expectations, rather than planning or perceptual distinctiveness. In duet performances, FRN was elicited in response to altered feedback not only for self-action but also partner-action, regardless of whether the performance was simultaneous (Loehr

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et al., 2013) or involved turn-taking (Huberth et al., 2019). As these piano performance studies only used prescribed musical scores, other negativities known for auditory expectation violation might have also contributed to the observed FRN, such as those based on transient auditory memory (mismatch negativity or MMN), tonal syntactic violations (early right-anterior negativity or ERAN), and conscious registration of task-relevant deviant-detection (N2b) (Folstein & Van Petten, 2008; Garrido et al., 2009; Horváth et al., 2008; Koelsch et al., 2000, 2003; Korka et al., 2022; Näätänen et al., 2007; Ritter et al., 1992). The current study further investigates improvisatory actions, which are not specifically bound to auditory memory, tonal knowledge, or deviant-detection requirements. Thus, we hope our study will add valuable insight to the related discussion.

It is important to note that action observation, another crucial part of joint action, is also involved in the performance monitoring process, as represented in the FRN/P3 complex. For example, the FRN is elicited in observers when others' actions result in unexpected feedback (Bellebaum & Colosio, 2014). The amplitudes of observers' FRN and P3 complex are generally lower than those of performers, suggesting that performers are more invested in processing the feedback of their own actions than observers (Koban et al., 2012). Using auditory feedback as the action-outcome modality, such as hearing a deviant pitch when one or another performer presses a piano key, gives similar results (Huberth et al., 2019; Loehr et al., 2015; Maidhof et al., 2010). These FRN and P3 complex responses are present when observing someone's musical action both as an audience (Maidhof et al., 2010) and as a coactor in a turn-taking task for tapping (Loehr et al., 2015) and for melody playing (Huberth et al., 2019). Specifically, Loehr et al. (2015) investigated performance monitoring in three settings: one's own individual performance, a partner's individual performance, and a joint performance where two participants alternate a tapping sequence. They found that the feedback in the joint performance elicited a larger FRN than when observing the partner's individual performance. This means that even in turn-taking when all actions momentarily belong to individual agents, the brain represents the task as jointly performed and evaluates the overall outcome.

In a joint task, the strength of action-outcome prediction is influenced by the similarity of the tasks performed by individuals. Coordination during joint task performance was higher if participants' motions shared similar spatial dynamic features (Słowiński et al., 2016). When pianists were asked to play with another audio-recording, they synchronized better with their own recording than with the recordings of others (Keller et al., 2007). Additional similarities like preferred tempo and prediction tendency between partners also enhanced temporal coordination in a joint performance (Loehr & Palmer, 2011; Pecenka & Keller, 2011). However, the effect of task similarity in a musical joint task on FRN/P3 responses may be more complex. Our previous piano-duet FRN-P3 study (Huberth et al., 2019) compared two conditions where duet scores used similar and dissimilar melodic contours between turn-taking duet parts. Unlike the findings regarding kinematic similarity, neither FRN nor P3 complex showed a significant difference. This raises the question of whether the strength of prediction may vary based on task similarity at a higher level than the representation of the melodic contour. The task similarity effects may come from the degree of understanding of the partners' task and its importance to the joint outcome.

Joint musical actions are also related to empathy. Empathy related to perspective-taking seems to enhance temporal interpersonal coordination. People with higher levels of perspective-taking scores anticipated tempo changes more accurately (Pecenka & Keller, 2011) and synchronized better with their partner in a joint musical coordination task (Novembre et al., 2019). This type of empathy is often measured by the perspective-taking subscale of the Interpersonal Reactivity Index (IRI) (Davis, 1980). However, in addition to perspective-taking, the Empathy Quotient (EQ, Baron-Cohen & Wheelwright, 2004) has been also shown relevant to coordination and representation of joint music performance. In our previous piano-duet work, pianists with a higher total EQ score showed greater adaptability in the keypress timing especially when duet melodic parts are dissimilar (Washburn et al., 2019) and exhibited reduced FRN in response to self-generated action (Huberth et al., 2019). Interestingly, EQ consists of subcategories of cognitive empathy, emotional reactivity, and social skills (Lawrence et al., 2004). Thus, empathy traits related to joint musical performance could be more complex than perspective-taking alone.

Further, it is important to note that the FRN magnitude was correlated with a variety of empathy scores in the literature; the total EQ score (Mothes et al., 2016), and both the IRI fantasy subscale (but not perspective taking) and the EQ emotional reactivity subscale (Fukushima & Hiraki, 2009). Moreover, the direction of FRN-EQ correlation varies depending on the task and agency. For example, a high EQ score resulted in more enhanced observational FRN in response to the rewards of the other agent (Fukushima & Hiraki, 2009), while a low EQ score was related to stronger FRN when unfairness was observed (Mothes et al., 2016) or when auditory feedback to the self-action was altered (Huberth et al., 2019). Studies involving nonmusical, social tasks demonstrated the effects of various other factors on FRN, such as reward, punishment, and fairness combined with gender and sociocultural contexts

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(Pfabigan, Wucherer, & Lamm, 2018; Pfabigan, Wucherer, Wang, et al., 2018) as well as individual traits like emotion (Starita et al., 2019), personality (Zhang et al., 2020), in addition to empathy (Fukushima & Hiraki, 2009; Mothes et al., 2016). Taken together, individual empathy traits might be related to the FRN in a more nuanced manner beyond perspective-taking only, potentially associated with multifaceted empathy in the EQ measures.

The present study aims to extend our understanding of performance monitoring of improvised actions in a joint task setting. Specifically, we extend our previous turn-taking scored piano-duet paradigm (Huberth et al., 2019) by introducing an improvisation task to examine how performance monitoring in the perception-action cycle is affected by improvisation, agency, and task similarity. Due to improvisation's real-time nature, it has been proposed that perception and action are dynamically coupled where action informs perception and vice versa (Clarke & Doffman, 2017). Thus, altered feedback during the improvisation may affect FRN and P3 complex differently compared to those when playing a musical score. In our task, a pair of pianists alternately played 6-note sequences (Figure 1) while their brain activity was recorded using EEG. Over multiple experimental blocks, the order of turn-taking between the partners and the type of performance task, score, or improvisation, were systematically changed to completely counterbalance the manipulations. In total, there were four distinct task combinations: (1) both pianists played the score, (2) the starting pianist played the score while the joining pianist improvised, (3) the starting pianist improvised while the joining pianist played the score, and (4) both pianists improvised. Task conditions were deemed "similar" when both players had the same assigned task (both playing the score or both improvising) and "different" when pianists had different tasks (one playing the score and the other improvising). For each trial in a block, the auditory feedback for each of the players was altered exactly once to another pitch (one that did not match the piano key pressed by the pianist) to elicit the FRN and P3 complex responses. The position of the altered feedback was also randomized at the phrase level and note position level such that neither pianist would be able to predict exactly when the feedback would be manipulated (see Section 2 and Figure 1 for details).

We investigated two competing hypotheses of neural responses to feedback errors in improvised and scored music. The first hypothesis postulates that the brain responses would reflect the prediction error and vary according to the strength of the prediction based on the encoded action plan. Since pianists play the same score repetitively over the course of the experiment, the scored melody is encoded as an action sequence in the brain more strongly than a melody that is created on the fly during improvisation. This hypothesis predicts that the response to the altered feedback in score-based performance will be larger than in improvised performance. Furthermore, this difference will be exaggerated for hearing altered feedback during one's own improvisation compared to a partner's improvisation. Predicting what the partner would play next is difficult while self-action might not be influenced by the distinction of whether the performance is based on a score or improvisation. Alternatively, our second hypothesis is that improvisation requires perception-action cycles to operate more dynamically, so the action sequence could be more tightly coupled with the feedback perception than in the repeated performance of scores. Thus, this alternate hypothesis predicts the opposite pattern of results regarding self-action feedback, where the FRN and P3 complex responses might be larger for improvisation than the score.

Further, we hypothesize that our turn-taking model would emphasize one's own action outcomes as more task-relevant than those of another's action, based on previous findings. This predicts that the FRN and the P3 complex would be larger for self-playing compared to a partner's playing. Furthermore, as demonstrated in previous work (Ullsperger et al., 2014; Walentowska et al., 2016), the FRN, P3a, and P3b responses reflect different functional stages of performance monitoring and outcome processing, namely, stimulus processing, attention orienting, and memory updating. Thus, these responses may exhibit variable sensitivities to agency and task similarity. Last, based on our and others' previous findings, we hypothesize that the joint performance monitoring process underlying the FRN for the self-action may be related to empathy measured by the EQ and their subscales.

2 | METHODS

2.1 | Participants

Twenty-six pianists, comprising 13 pairs, were recruited for the study. They were 21–42 years old ($M = 26.3$, $SD = 4.5$) and had 8–23 years of general musical experience ($M = 16.2$, $SD = 4.2$). Fourteen subjects had 2–20 years of improvisatory musical experience ($M = 8.8$, $SD = 5.4$). Twenty-five pianists were self-reportedly right-handed and one was ambidextrous. We tested a total of 26 pianists and included all in our final ERP analysis. However, behavioral data logging for keyboard timing and sound failed in one duet pair. Thus, the behavioral analysis involved the remaining 24 pianists, including 13 pianists with improvisation experience ($M = 9.2$, $SD = 5.4$ in years). The study protocol was approved by Stanford University's Internal Review Board. The pianists provided written

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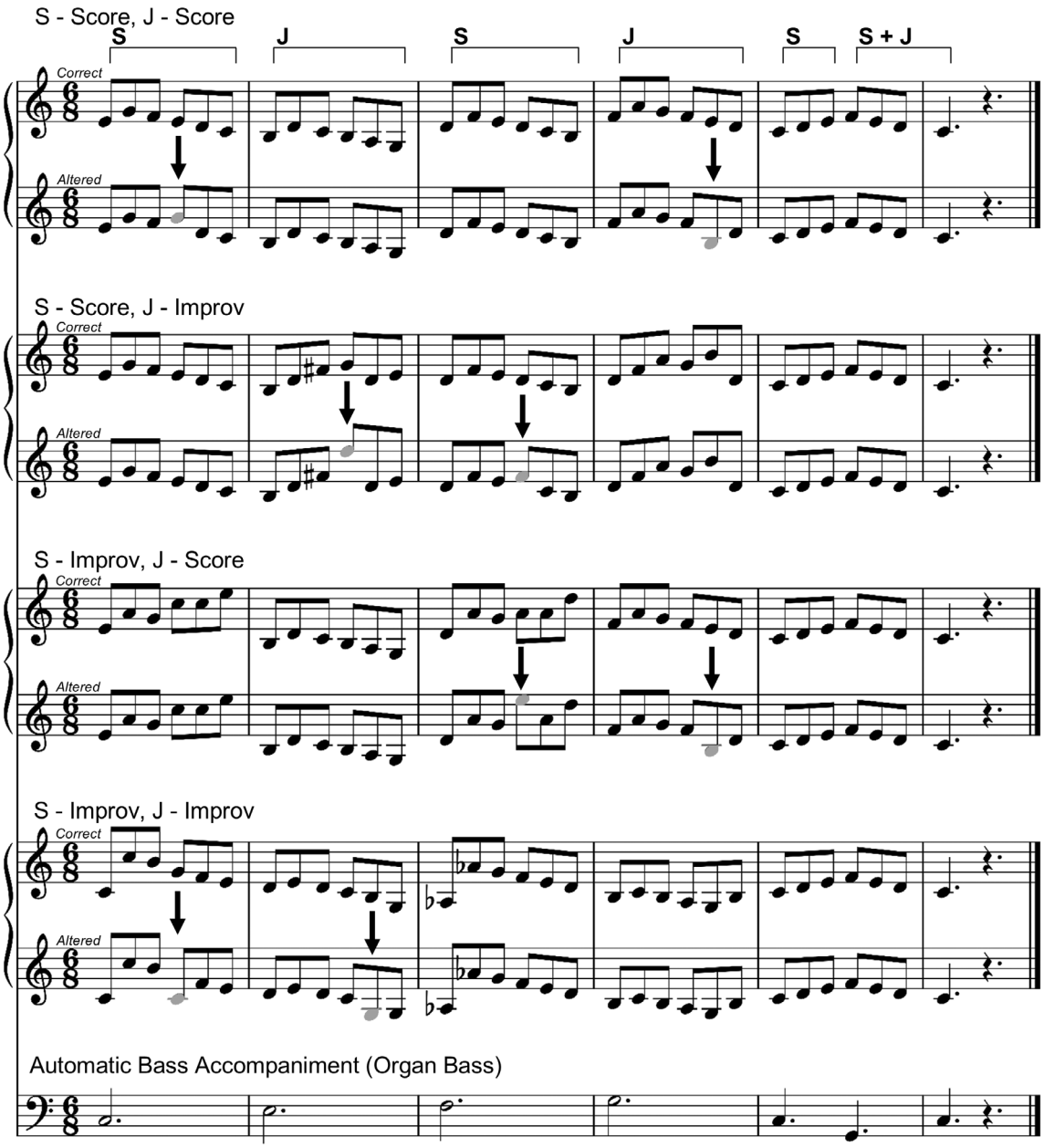

FIGURE 1 Examples of the four duet types played by two pianists (S—Starter, J—Joiner) and altered auditory feedback. The first type shows the fixed score used for all Score conditions, whereas the other three types shown here include examples of improvised melodies. In every condition, the players alternated measures during the first four measures (1(S)-2(J)-3(S)-4(J)), and then the Joiner joined the Starter for the four unison ending notes in measures 5 and 6. These ending notes in the measures 5 and 6 were the same for all trials and duet types, even when improvising. Altered pitches (colored in gray) occurred once per pianist per trial, during the first four measures, changing the auditory feedback to two, three, or four scale tones above or below the player's input. The bottom line shows the bass accompaniment, which was automatically played by the computer program and synchronized with pianists throughout the trials to provide harmonic context and aid players' improvisatory creativity.

informed consent forms and received a gift card worth the rate of $20 per hour.

Our sample of 26 pianists in the final ERP analysis exceeded the sample size of 18 in our previous piano-duet FRN/P3 study (Huberth et al., 2019). We calculated the required sample size using their results in the main effect of *Agency* (self vs. other), the only common main effect. In their results, the main effect of *Agency* for FRN was significant with an F-value ($F(1, 17)$) of 33.32 and η^2 of 0.21. That for P3a was significant with an F-value ($F(1, 17)$) of 11.12 and η^2 of 0.08. With a repeated-measures design for this factor (2 levels, 4 measurements) and Cohen's f derived from these η^2 values, we obtained a minimum sample size of $N=8$ for FRN and $N=18$ for P3a with the significance

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level α of .05 and the power of 0.8 (G*Power ver.3.1.9.7, Faul et al., 2007). Thus, our final sample size satisfies the minimum sample size for both FRN and P3a.

2.2 | Task and apparatus

Two participants sat face-to-face in an acoustically and electrically isolated room, each in front of an M-Audio Axiom-61 digital keyboard. The keyboards were connected to custom software implemented in Max/MSP (7.0.1, Cycling '74, Covina, CA) running on a Macintosh computer (OSX 10.9.5), which monitored the timing and MIDI pitch numbers of key presses according to the score (see below for details). The software also controlled metronome count-in ticks, bass accompaniment, and the piano-timbre feedback (built-in sound of AU DLS Synth). The sound was presented through two loudspeakers placed on both sides of the keyboards.

Each pianist spent half of the recording time playing precomposed sequences indicated in Figure 1, and the other half playing improvised sequences, resulting in four different combinations of duet task types between self and partner (score + score, score + improvisation, improvisation + score, and improvisation + improvisation). Players strictly followed the score whenever they were assigned to the score condition. As shown in the score, this 31-note piano duet consisted of 6 measures in a 6/8 time signature and was specifically composed for the study by the authors. For the first 4 measures, each player alternated each measure consisting of 6 isochronous notes. After these alternating four measures, came the 5th and 6th measures, where the starting player (henceforth called "Starter") played the first 3 notes and was joined by the following player ("Joiner") for the remaining four notes. Thus, in each trial, the Starter played the first and third measures and the Joiner played the second and fourth, resulting in a turn-taking musical interaction. When either pianist improvised, they had to play 6 notes in a strict isochronous eighth-note rhythm (comprising one measure) with any pitch content for their first two turns. However, they were instructed to play the fifth and sixth measures as written in the score regardless of assignment to score or improvisation tasks. Across all conditions, pianists played the keyboard by their right hand only.

Performances were accompanied by an organ bass part (Figure 1), which sustained a single low-pitched note through each measure so that pianists could use its harmonic context to aid improvisatory creativity. The bass sequence was crafted based on a conventional harmonic progression (Tonic—Subdominant—Dominant—Tonic). The bass notes were fixed for both score and improvisation to maintain the balance of the tonal context to the original key. They were controlled by the same Max/MSP program so that it synchronized in time to the players' performance. At the start of each measure, an organ note would start exactly when the first note of the measure was performed by the designated pianist and was sustained until the first note of the following measure or, in the fifth measure, the first note of the next harmonic context. In the fifth measure, the fourth note of the measure (with a corresponding change in bass note) was played by both Starter and Joiner, who in general would not be perfectly synchronized; in this case, the earlier keypress for that note position would trigger the organ bass note.

Altered pitch feedback for each pianist was designed to occur either at the fourth or fifth note of one of the first four measures (Figure 1). The altered feedback was introduced once per trial in one of the two measures for each pianist, and the probability of the alteration for the measures and note positions was equally distributed such that neither pianist could predict which note would be altered in any given trial. Altered notes were always in-key (from the same C major scale as the scores) and were two, three, or four diatonic scale notes above or below the actual key pressed by the player. There were no restrictions placed on which keys the pianists could play for improvisation and they were allowed to play black keys which were not part of the original score written in C major. Even in those cases, however, the altered feedback was given with an in-key pitch (e.g., a tone belonging to a white key) as the bass notes provided corresponding harmonic context to the key and to avoid potential confounds of out-of-key pitch processing from the target ERP components. For convenience and simplicity going forward, a correct pitch in these positions is called *Standard*, whereas an altered pitch is called *Deviant*.

A block consisted of 16 trials preceded by two practice trials, and each trial began with three isochronous metronome "count-in" clicks with a 500-ms interonset interval (IOI) to set the tempo and inform the participants to start the trial. After completing a previous trial, there was a jittered silence (1.5–2.5 s duration) between the onset of the final note and the first metronome click of the subsequent trial. Pianists were instructed to strictly maintain the rhythm in the notated sequences in both the score and improvisation conditions. Also, pianists were made aware of the presence of altered feedback to one of their phrases every trial. They were further instructed not to react and continue playing in order to complete a trial successfully.

Our Max/MSP program monitored both pianists' accuracy in rhythm and pitch, inserted deviant pitches, and handled performance errors in real-time while keeping complete log files of time-stamped events such as keypress, metronome click, and trial outcome. A trial was regarded as unsuccessful if pianists played out of turn,

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pressed a wrong key in scored sequences, or if the IOI was 125 ms above or below the expected IOI (500 ms). When a trial was deemed unsuccessful, the same trial type (with regards to the positions and the pitch intervals of the deviant notes) was appended to the end of each block and had to be successfully performed. Pianists had to play both practice trials, which contained no deviant pitches, correctly to proceed to the test trials. In the test trials, pitch alterations occurred only once in each player's part per trial; as a result, each player played one measure containing a deviant and another containing no deviant.

The Max/MSP software also provided trigger codes to the parallel port of the EEG acquisition system through a custom USB-to-parallel system based on Arduino Uno. The different trigger codes were designed to express information regarding the trial duet type, metronome clicks, the keys pressed by the pianists through MIDI inputs, standard vs. deviant notes, the pitches those keypresses produced, as well as the success or failure at the end of each trial. The latency of the trigger timepoint and the audio playback was 21 ms as in our previous setup (Huberth et al., 2019). We compensated for this latency in the EEG analysis (see below Section 2.4).

The recording session started with one pianist performing the Starter role and the other performing the Joiner role, for each of the four duet types. Then pianists switched roles, going through the four duet types in the same order, without switching the keyboard they were assigned to. These eight blocks were called a superblock. This superblock was then repeated three times throughout the session for a given pair, resulting in 48 test trials for each of the eight combinations of individual role and duet type. A session took approximately 4 hours total including instruction, EEG cap outfitting, and breaks. The order of the four duet types was counterbalanced across different pairs in the study.

After the EEG session, pianists were asked to fill out a questionnaire regarding demographic and musical background and the 40-item Empathy Quotient (Baron-Cohen & Wheelwright, 2004).

2.3 | Behavioral data analysis

Every note played by the participants in the experiment as well as auditory feedback was recorded using MIDI numbers and stored as the experiment session was administered in Max/MSP. We organized the data file based on block, trial, condition, and order of notes played and used that information to perform three behavioral analyses.

First, we investigated how the participants improvised, by analyzing the number of unique melodies played per block (max 32, as 2 phrases are played in each trial where a block contained 16 trials), the number of unique notes played per melody (max 6), and the number of notes outside the C major scale also played per melody (max 6). One-sample *t*-tests were conducted for within-subject comparison of these measures based on whether the partner is playing the score or improvising jointly. Additionally, a two-sample *t*-test compared subject groups such as those with or without improvisation experience on the creativity measures.

Second, we analyzed the melodic correlation values of an improvised phrase to the corresponding part of the score, the other phrase improvised by the same pianist within the same trial, and the other pianist's phrase preceding or following the phrase. The correlation between two 6-note melodies was computed in the following steps: (1) Transform each melody as a series of 6 MIDI numbers to a series of 5 absolute interval values between two consecutive notes (e.g., each corresponding to the number of semitone differences). (2) Add an additional 0.1 to each interval value to avoid zero as an interval for the repeated notes. (3) Calculate the ratio of two interval values from melody #1 and melody #2 for each of the five positions while making sure the larger interval value becomes the denominator. (4) Average of the five ratio values. This results in a number between 0 and 1.0 (while the minimum correlation would not be 0, the maximum correlation of 1 would be obtained from two completely transposed melodies). For the correlation, we used the MIDI numbers from keypress for the self-improvised melodies to account for the player's intention. In contrast, for the phrase from the partner, we used MIDI numbers from auditory feedback, including altered feedback at the deviant position, to account for the player's perception of the "heard" melody rather than the "played" melody. We used paired *t*-tests to compare the correlation values.

Third, using a two-way repeated measures analysis of variance (ANOVA), we compared IOI differences between IOIs of a target note and its neighboring notes based on the task type (score vs. improvisation) and the phrase type (deviant-phrase vs. standard-phrase) to examine whether the altered feedback disrupted the pianist's key press timing. The positive/negative IOI difference indicates slowing/speeding for the note right after the target position compared to the note before.

The 40-item Empathy Quotient questionnaires were scored using the methods described in Baron-Cohen and Wheelwright (2004). Then the scores were separated into subscales of cognitive empathy, emotional reactivity, and social skills based on Lawrence et al. (2004), which used principal component analysis and final loadings higher than 0.3 for each component. Therefore, in our analysis, a total of 13 questions (Q1, Q19, Q22, Q25, Q26, Q36, Q41, Q43, Q44, Q52, Q54, Q55, and Q58) were used for the

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cognitive empathy subscore, 12 for emotional reactivity (Q6, Q21, Q22, Q27, Q29, Q32, Q36, Q42, Q43, Q48, Q50, and Q59), and 7 for social skills (Q1, Q4, Q8, Q12, Q14, Q35, and Q57).

2.4 | EEG recording and analysis

The EEG of both pianists were recorded simultaneously at a 500 Hz sampling rate using the Neuroscan SynAmpsRT amplifier with Curry 7 acquisition software and two whole-head 64-channel Quikcaps (all three from Compumedics, El Paso, TX). EEG data of both participants were recorded simultaneously and saved in one file. Electrode impedances were kept below 10 kΩ throughout the experiment. The horizontal and vertical electrooculogram (EOG) was also recorded for each participant. Offline, the files were separated for each participant and re-referenced using the common average and the resulting data underwent DC-drift compensation. The EOGs were then used to remove eye artifacts using Source Space Projection implemented in the Brainstorm toolbox (Tadel et al., 2011) in MATLAB (Mathworks Inc., Natick, MA). We used a single, continuous raw file to detect typical eye artifacts (blinks and horizontal movements) and used these events to construct topographic maps called "projectors" via principal component analysis provided by the Brainstorm toolbox. The set of these artifact-related projectors for each participant was subtracted from each epoch of 4000 ms, beginning 2000 ms before the onset of the fourth and fifth note of each of the first four measures in successfully completed trials. Epochs were downsampled to 125 Hz. Epochs with peak-to-peak amplitudes outside ±150 μV on any channels were rejected.

The EEG conditions were distinguished based on four features: whether the pitch of the note (fourth and fifth notes of first–fourth measures) was altered or not (Standard vs. Deviant), whether the note was played by the player or their partner (Self vs. Other), whether the Deviant note occurred during score or improvisation (Score vs. Improv), and whether the partners' tasks in that block were the same (both playing score or both improvising) or different (one playing score while the other improvises; Same vs. Different). Within each participant, all instances of these conditions throughout the experiment were collected and averaged. The 21-ms latency of the trigger timing to the onset of auditory feedback (piano-tone playback) was corrected at this point. Then, baseline correction was performed using a 100 ms prestimulus window.

We calculated difference waves for each average by subtracting the ERPs in response to Standards from those in response to Deviants. Next, we averaged the ERP data and the difference waveforms into two large electrode groups: frontocentral group (F1, Fz, F2, FC1, FCz, and FC2) for FRN and P3a (see Figure 3 inlay) and parietal group (CP1, CPz, CP2, CP4, P1, Pz, P2, and P4) for P3b (see Figure 4 inlay). Both topographies showed the frontocentral concentration for FRN and P3a, and the centroparietal concentration for P3b, largely consistent with previous findings (Ullsperger et al., 2014).

To examine the amplitude of FRN, P3a, and P3b, we established a time window of interest for each component using the method for selecting a window without bias as recommended by Keil et al. (2014). First, we averaged the difference waveforms across all conditions, roles, and participants at the two electrode groups. After identifying a peak of interest (FRN/P3a/P3b), two time-points around the grand-average peak corresponding to 50% of that peak's amplitude were identified (full width at half maximum). Through this process, we obtained a time window of 131–187 ms for FRN (peak at 163 ms), 219–291 ms for P3a (peak at 251 ms) using the frontocentral site, and 427–579 ms for P3b (peak at 475 ms) using the centroparietal site.

For statistical analysis of these amplitude values, we used repeated measures ANOVAs with three within-participants factors (*Type* [Score, Improv], *Agency* [Self, Other], and *Similarity* [Same, Diff]). Note that the factor type corresponds to the task type of whichever pianist played the Deviant note. To measure the effect size for ANOVAs, we report generalized eta squared, η_G^2(Bakeman, 2005). For time-window calculation and visualizations of grand average ERPs and their topographies, we bandpass-filtered (0.1–40 Hz) the data, but for statistical analysis, we used the unfiltered data.

3 | RESULTS

3.1 | Behavioral data for improvisation

Table 1 shows that pianists (1) played a novel melody almost every time during a block out of 32 opportunities, (2) only used about one repeated note within a 6-note phrase, and (3) seldom used out-of-key notes. There were no significant differences in self-improvisation when comparing whether the partner was improvising or playing the score for any of the measures: the number of unique melodies ($p = .065$), the number of unique notes ($p = .56$), and the number of out-of-key notes ($p = .99$) played by a participant.

The results in Figure 2a show a significant correlation difference based on whether the partner was improvising or playing the score. Paired *t*-tests show that the correlation between the two improvised phrases played

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TABLE 1 Improvisation performance separated by partner's task types.

	Number of unique melodies per block (max 32)		Number of unique notes per melody (max 6)		Number of out-of-key notes per melody (max 6)	
	M	*SD*	*M*	*SD*	*M*	*SD*
Partner is improvising	31.28	1.04	4.6	0.29	0.82	0.59
Partner is playing the score	30.98	1.36	4.66	0.32	0.82	0.70

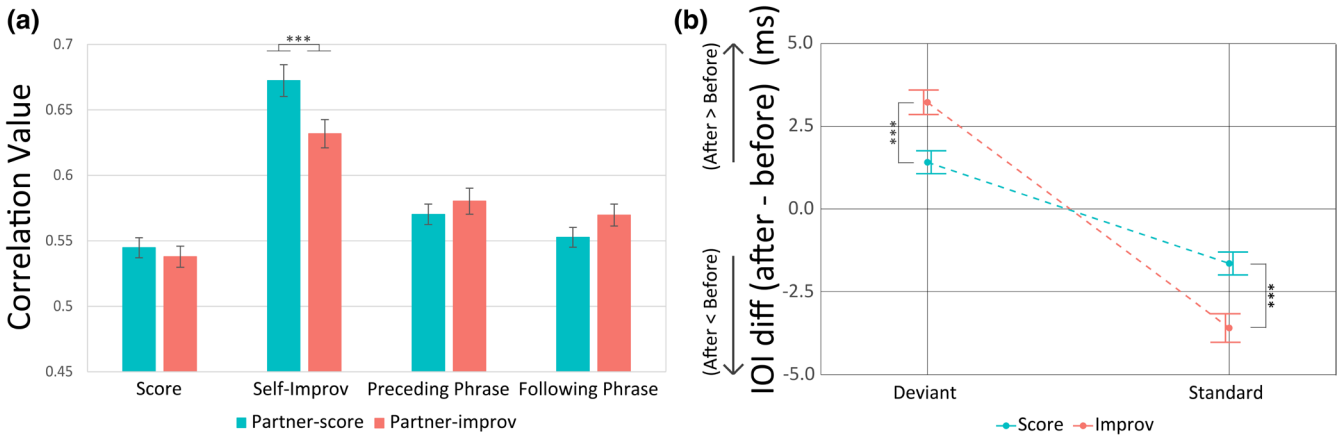

FIGURE 2 (a) Behavioral results on the correlation of melodies. Semitone intervals between two successive notes in a 6-note melody were used for melodic contour correlation analysis (explained in detail in the Section 2). Four different correlation values of a given improvised phrase are computed with (1) Score (the score phrase of the corresponding measure), (2) Self-Improv (the other self-improvised phrase within the trial), (3) Preceding Phrase (partner's phrase heard before), and (4) Following Phrase (partner's phrase heard after). The correlation values were compared based on partner's task types (Partner-score and Partner-improv mean partner is playing the score or improvisation, respectively). (b) Inter-onset interval (IOI) differences between before and after a target note (after minus before). Within a given phrase, the target note is defined as the fourth or fifth note of a phrase. The IOI differences were separated into the feedback type of the target note (Deviant or Standard) and task type (Improvisation or Score). The error bar expresses the standard error of the mean (SEM). (*** denotes $p < .001$).

by the same pianist within a trial was stronger when the partner was playing the score compared to improvising ($t(23) = 4.94$, $p < .001$). There was no significant difference in the correlation of self-improvisation to score ($t(23) = 1.30$, $p = .21$) and to the partner's phrase heard before ($t(23) = -0.86$, $p = .40$) or after ($t(23) = -1.79$, $p = .086$) based on the partner's task type. As seen in Figure 2a, within-subject comparisons revealed that the self-improvisation showed the highest correlation among the four correlation types, significantly higher compared to the original score ($t(23) = 9.04$, $p < .001$), the preceding ($t(23) = 8.38$, $p < .001$), or the following melodies ($t(23) = 8.81$, $p < .001$).

IOI differences between before and after the target note in a phrase are shown in Figure 2b. The ANOVA revealed a significant main effect of *Feedback* [Deviant, Standard] ($F(1, 23) = 22.11$, $p < .001$, $\eta_G^2 = 0.25$). The ANOVA also revealed a significant two-way interaction between *Feedback* and *Type* [Improv, Score] ($F(1, 23) = 5.81$, $p = .024$, $\eta_G^2 = 0.041$). *Post hoc* comparisons showed that when the pianists were improvising at the target position, they slowed down more after hearing altered feedback ($p < .001$), whereas they rushed more in the absence of altered feedback ($p < .001$) compared to the score condition. No significant main effect or interactions were revealed for *Similarity* [Same, Different].

3.2 | Event-related potentials

We investigated whether our data showed FRN, P3a, and P3b ERP components. Figure 3a shows the time-series plots in the frontocentral area and topography maps at the time of FRN and P3a peaks. Responses to altered pitches compared to the unaltered pitches showed negative deflections around 160 ms and positive deflections around 250 ms on frontocentral scalp distributions. Mean amplitudes of the difference waveforms for FRN and P3a are shown in bar plots (Figure 3b). Individual amplitude values were assessed by repeated measures ANOVA using

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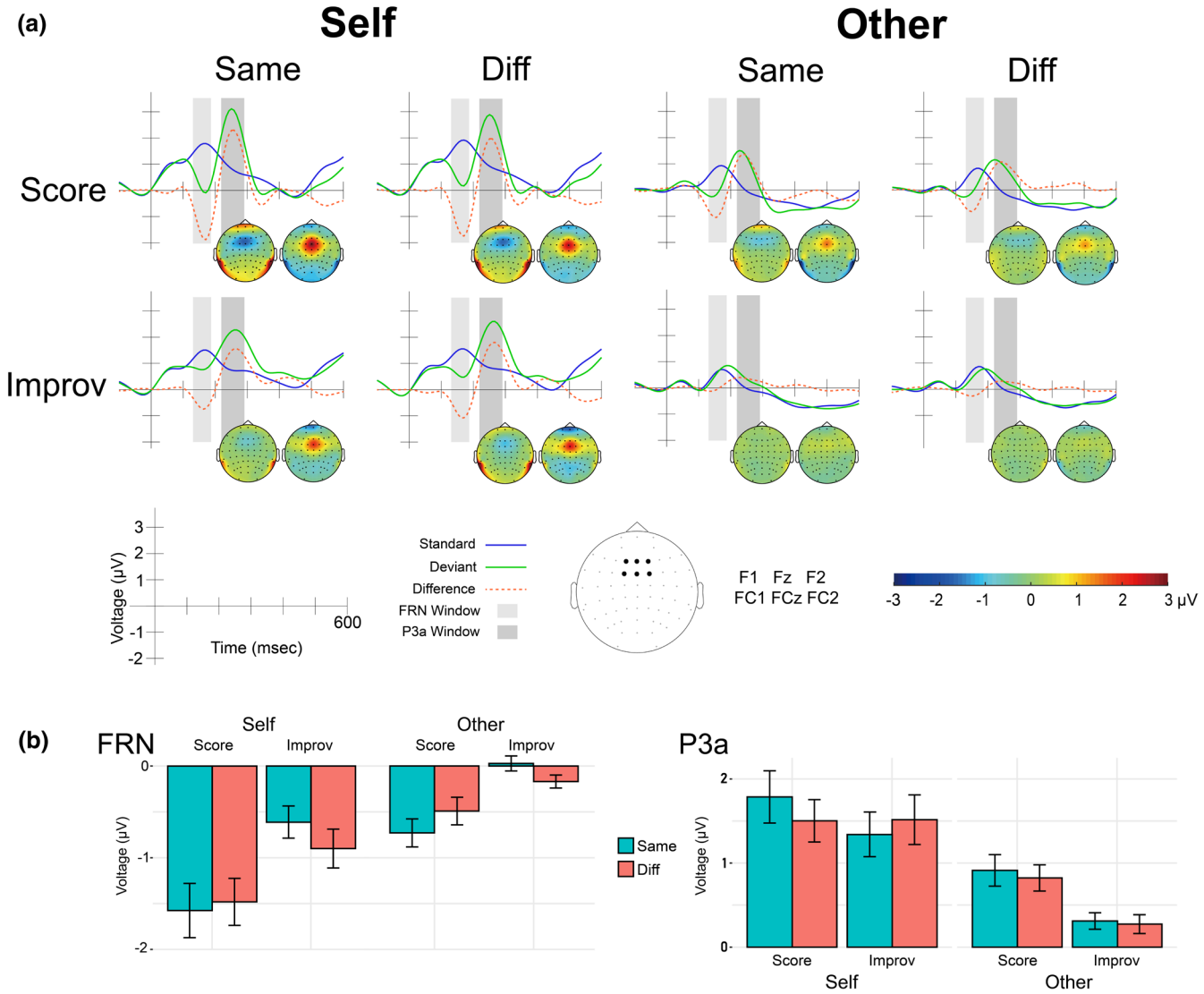

FIGURE 3 (a) Grand average time series plot of frontocentral electrode group (F1, Fz, F2, FC1, FCz, and FC2) and topographies for each condition. The ERPs for Standard and Deviant feedback are indicated with solid blue and green lines, respectively, whereas the difference waveforms (Deviant minus Standard) are indicated with red dashed lines. Topographies of FRN (left) and P3a (right) are shown in each condition, obtained from the time windows indicated with light and dark gray colors in the time series, respectively. (b) Mean amplitude for FRN (left) and P3a (right). The error bar expresses the SEM.

three within-participant factors (*Type* [Score, Improv], *Agency* [Self, Other], and *Similarity* [Same, Diff]).

For FRN, the ANOVA revealed a significant main effect of *Type* ($F(1, 25) = 32.22$, $p < .001$, $\eta_G^2 = 0.11$), reflecting that altered feedback on scored melodies elicited larger FRN (i.e., more negative) than on improvised melodies. The ANOVA also revealed a significant main effect of *Agency* ($F(1, 25) = 27.53$, $p < .001$, $\eta_G^2 = 0.15$), reflecting that FRN was larger for one's own part than for the partner's part. No significant main effect of *Similarity* was observed. However, the ANOVA revealed a two-way interaction between *Type* and *Similarity* ($F(1, 25) = 11.00$, $p < .005$, $\eta_G^2 = 0.012$). *Post hoc* comparisons revealed that the FRN for scored melodies was larger when both pianists played the score compared to when one of them played improvised melodies ($p = .041$). Similarly, the FRN was larger for improvised melodies when one of the pianists played the score compared to when both played improvisation ($p < .005$).

For P3a, the ANOVA revealed a significant main effect of *Type* ($F(1, 25) = 16.01$, $p < .001$, $\eta_G^2 = 0.031$), reflecting that the altered feedback on scored melodies elicited larger P3a than on improvised melodies. The ANOVA also revealed a significant main effect of *Agency* ($F(1, 25) = 24.80$, $p < .001$, $\eta_G^2 = 0.16$), reflecting that the altered feedback for one's own part elicited larger P3a than for the partner's part. No significant main effect or interactions of *Similarity* were observed.

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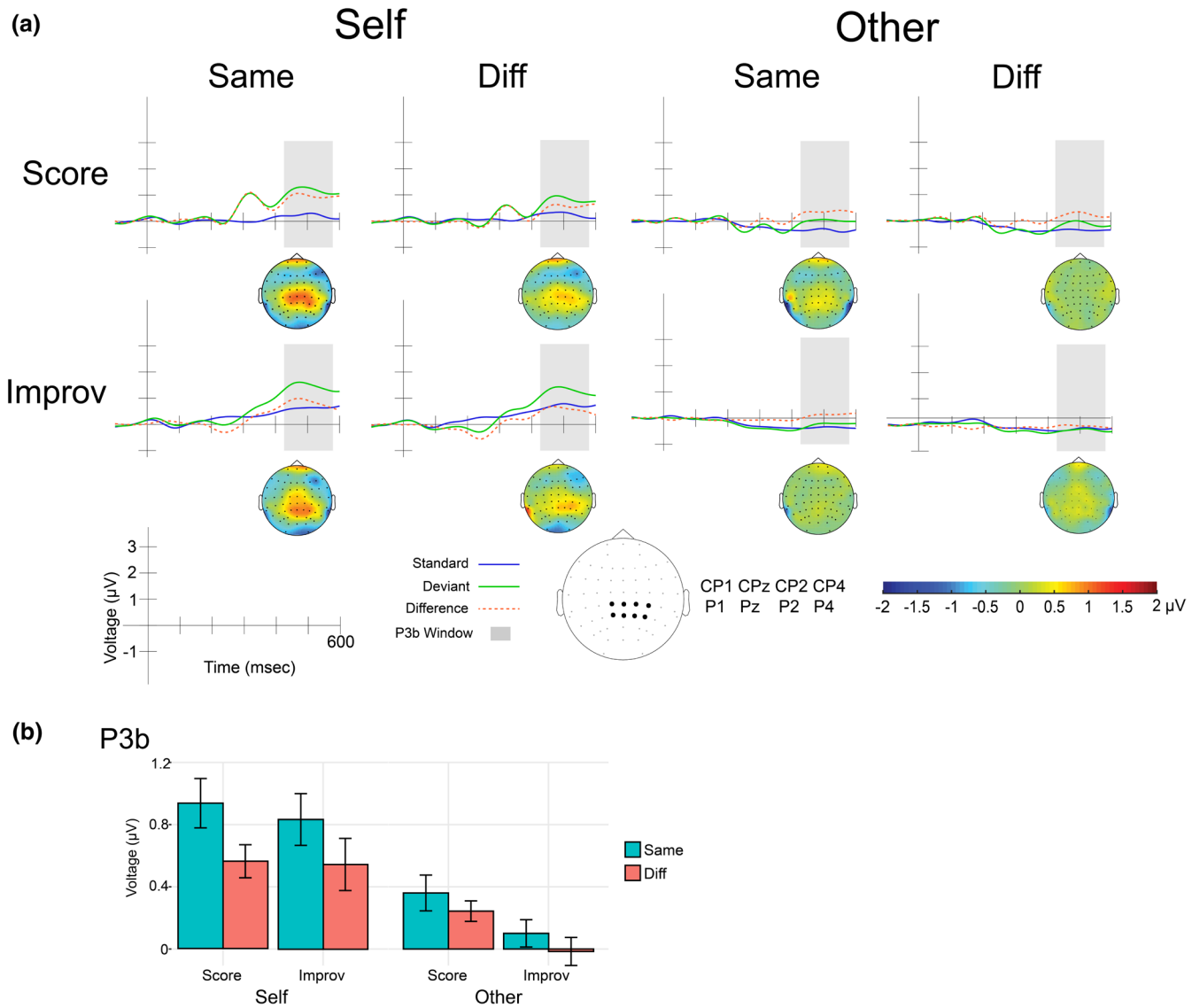

FIGURE 4 (a) Grand average time series plot of parietal electrode group (CP1, CPz, CP2, CP4, P1, Pz, P2, and P4) and topographies of P3b for each condition, obtained in the time window indicated by the light gray color in the time series. (b) Mean amplitude for P3b. The error bar expresses the SEM.

Figure 4 shows the time-series plots in the centroparietal area and the topography maps at the time of P3b peaks. Response to altered pitches compared to correct pitches showed a sustained parietal wave, beginning at approximately 430 ms. Again, mean amplitudes of the difference waveforms regarding P3b were entered into repeated measure ANOVA using three within-participants factors (*Type* [Score, Improv], *Agency* [Self, Other], and *Similarity* [Same, Diff]).

For P3b, the ANOVA revealed an effect of *Agency* ($F(1, 25) = 26.78$, $p < .001$, $\eta_G^2 = 0.16$), reflecting that the P3b for a player's own part was larger than for their partner's part. In addition, the ANOVA revealed an effect of *Similarity* ($F(1, 25) = 7.12$, $p = .013$, $\eta_G^2 = 0.031$), reflecting that P3b was larger when both players played similar types of melodies (Scored or Improvised) than when the two players followed two different strategies. There were no other effects or interactions. For the complete ANOVA results for the ERP data, see Supplementary Material Section 1, Table S1.

Finally, because these three components were relatively small for a partner's improvisation, we tested whether the response amplitude was different from zero (i.e., pre-stimulus baseline) by a two-tailed, one-sample *t*-test. The FRN amplitude was significantly less than zero when the participant was playing the score ($t(25) = 2.39$, $p = .025$), but not when the participant's own task was to improvise ($t(25) = 0.34$, $p = .74$). The P3a amplitude was significantly greater than zero regardless of the self-task context ($t(25) = 3.96$, $p < .001$). The P3b was not significantly different from zero.

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3.3 | Correlations to empathy

We investigated the relationship between the players' trait empathy, as measured by the Empathy Quotient ($M = 42.7$, $SD = 12.08$), and the FRN amplitude using Spearman rank-order correlations. Correlations using subscores of empathy (cognitive empathy, emotional reactivity, and social skills) revealed marginal significance between emotional reactivity (12 items: $M = 14.27$, $SD = 4.27$) and FRN amplitude in response to self-played phrases ($r = .46$, $p = .019$, uncorrected), which was not significant after Holm-Bonferroni correction for multiple testing involving three subscores ($\alpha = .017$). The detailed results and associated discussion are included and reported in Supplementary Material Section 4 and Figure S1.

3.4 | Effects of improvisation experience on behavioral and ERP data

Table 2 shows behavioral data separated by improvisation experience. There were no significant differences in self-improvisation based on improvisation experience in the number of unique melodies ($p = .47$), the number of unique notes in a phrase ($p = .84$), and the number of out-of-key notes per melody ($p = .38$) played by a participant.

IOI differences (after minus before the target note in a phrase) are shown in Figure 5a for experienced and inexperienced improvisers. As for the IOI differences, we examined a mixed-design ANOVA with a between-subject factor *Expertise* (experienced, inexperienced) in addition to within-subject factors *Feedback* [Deviant, Standard] and *Type* [Improv, Score]. Three-way *Expertise* × *Feedback* × *Type* interaction was marginal ($F(1, 22) = 4.20$, $p = .052$, $\eta_G^2 = 0.0018$) (for the complete ANOVA results, see Supplementary Material Section 2). Following this interaction, separate ANOVAs were conducted for each Expertise group. The ANOVA for experienced improvisers revealed a significant main effect of *Feedback* ($F(1, 12) = 10.38$, $p < .01$, $\eta_G^2 = 0.25$) and a significant two-way interaction between *Feedback* and *Type* ($F(1, 12) = 9.87$, $p < .01$, $\eta_G^2 = 0.13$). *Post hoc* comparisons showed that when the experienced improvisers were improvising, they slowed down more after hearing altered feedback at the target position ($p < .001$), whereas they tended to rush more in the absence of altered feedback ($p < .001$). In addition, post-deviant slowing was greater for improvisation than for score ($p < .001$). The ANOVA for inexperienced improvisers revealed a significant main effect of *Feedback* ($F(1, 10) = 10.98$, $p < .01$, $\eta_G^2 = 0.28$). No other significant main effects or interactions were revealed for inexperienced improvisers.

As for the ERP amplitudes, similarly, we used a mixed-design ANOVA with the between-subject factor of *Expertise* (experienced, inexperienced) with within-subject factors *Type*, *Similarity*, and *Agency*. The *Expertise* x *Type* interaction was significant in P3a amplitude ($F(1, 24) = 5.36$, $p = .029$, $\eta_G^2 = 0.0092$), but no main effects or interactions were found for FRN and P3b (for the complete ANOVA results, see S1, Table S3). We further examined P3a amplitudes separately for experienced and inexperienced improvisers, as shown in Figure 5b. The ANOVA for experienced improvisers revealed a significant main effect of *Agency* ($F(1, 13) = 13.30$, $p < .005$, $\eta_G^2 = 0.23$) and a two-way interaction between *Type* and *Agency* ($F(1, 13) = 6.33$, $p = .026$, $\eta_G^2 = 0.022$). The ANOVA for inexperienced improvisers revealed a significant main effect of *Agency* ($F(1, 11) = 10.56$, $p < .01$, $\eta_G^2 = 0.11$) and *Type* ($F(1, 11) = 15.32$, $p < .005$, $\eta_G^2 = 0.056$). *Post hoc* comparisons showed that the P3a difference between self-score and self-improv was not significant for experienced improvisers ($p = .63$) but was significant for inexperienced improvisers ($p = .022$).

4 | DISCUSSION

The present study investigated ERPs pertinent to performance monitoring in a turn-taking piano duet where two pianists played either scored or improvised melodies. Our key findings are as follows. (1) Altered pitch feedback during the self-action elicited FRN and P3 complex regardless of playing the score or improvising, although the FRN and behavioral response differed

TABLE 2 Improvisation performance separated by improvisation experience.

	Number of unique melodies per block (max 32)		Number of unique notes per melody (max 6)		Number of out-of-key notes per melody (max 6)	
	M	*SD*	*M*	*SD*	*M*	*SD*
Has improvisation experience ($N = 13$)	31.29	1.19	4.63	0.27	0.93	0.57
Has no improvisation experience ($N = 11$)	30.94	1.14	4.66	0.31	0.70	0.69

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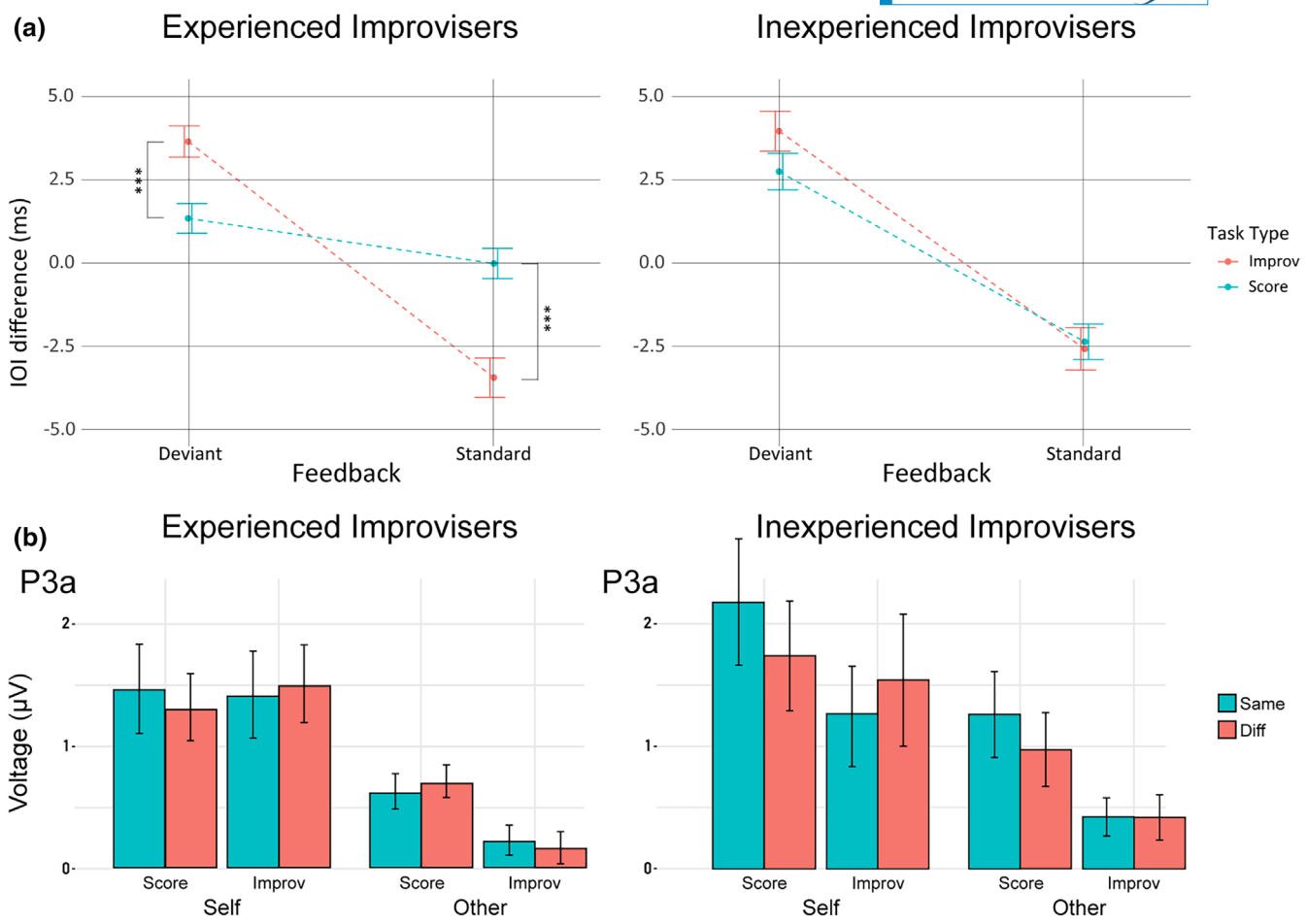

FIGURE 5 Inter-onset interval (IOI) differences and P3a amplitudes separated into two groups: Experienced improvisers and inexperienced improvisers. (a) IOI differences between before and after a target note (after minus before). Within a given phrase, the target note is defined as the 4th or 5th note of a phrase. The IOI differences were divided into the feedback type of the target note (Deviant or Standard) and task type (Improvisation or Score). (b) Mean amplitude for P3a. The error bars of (a) and (b) express the SEM. (*** denotes $p < .001$).

between scored and improvised melodies. (2) The FRN and P3 complex were enhanced for self-generated melodies compared to partner-generated melodies. (3) The between-partner task similarity affected the FRN and P3b, whereas score vs. improvisation difference did not matter for the latter.

The FRN and P3 complex were observed for both score and improvisation during self-action, which was in line with previous studies involving feedback manipulations during music performances (Huberth et al., 2019; Katahira et al., 2008; Loehr et al., 2013; Maidhof et al., 2010; Mathias et al., 2019). Most notably, the novel aspect of our finding is that the FRN/P3 were clearly present during self-improvisation, extending previous FRN findings using only a score. The observed FRN during improvisation shows that action–perception coupling works dynamically even when pianists were generating a novel melody almost every time. While generating variability is a crucial aspect of improvisation tasks, potentially attenuating the FRN with reduced expectations of tone alterations (Hajcak et al., 2005, 2007), this did not abolish the FRN.

Still, our data revealed that players exhibited a greater magnitude of feedback processing when playing or hearing scored melodies than improvised melodies. Importantly, this pattern holds true both during self-generated actions, and during partner-generated actions. This supports the account that prediction is stronger for a preconceived action than an improvised one. When score materials are played repetitively, as in our case, the action sequence becomes well-rehearsed and memorized. Repeated tasks may also provide ample motor simulation, enhancing the degree of action prediction (Candidi et al., 2014; Novembre et al., 2014). These findings are consistent with an internal forward model (Martikainen et al., 2005; Wolpert et al., 1995), where well-rehearsed actions would produce stronger predictions for both action outcomes and enhanced monitoring processes. Improvised melodies, in contrast,

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differ in contour, pitch, and pattern every time, likely leading to weaker encoding and prediction processes. Interestingly, unlike self-generated FRN, self-generated P3a and P3b differences between score and improvisation were not significant. Our instruction to ignore altered pitches and continue playing was the same for both the score and improvisation conditions. Therefore, having no difference in task relevance and memory updates (reflected by the P3 complex) may be the reason behind the absence of significant differences.

Improvisation may have affected performance monitoring differently from score-playing in addition to the repetition-related action/auditory familiarity. Our behavioral results in the keypress timing patterns showed a sharp contrast between the score and improvisation, not only for altered feedback but also for standard feedback. The pianists slowed down after hearing altered feedback on the target position (fourth or fifth note of a measure), with a greater delay during improvisation compared to score-playing. However, they rushed after hearing standard feedback on the target position, more pronouncedly during improvisation. Thus, pianists' approach, goal, and cognitive workload when improvising may be different from score-playing, reflected in phrase-level action–perception coupling. In fact, previous findings support this notion such that action–perception coupling exists in a multi-layered manner with individual pitch-to-action mapping and higher sequence-level melodic information. For example, in one study, the premotor cortex was activated when nonmusicians were listening to a previously trained melody, but not when they were listening to an untrained melody made by reordering the same set of finger-note combinations (Lahav et al., 2007). Implicit transfer from individual pitch-action mappings to a sequence-level representation of novel melodies has also been found, reflected by the activation in the left dorsal premotor area (de Manzano et al., 2020). In addition, interference over the dorsal premotor area impeded learning novel sequences with scrambled notes, but not when the notes were simple ascending/descending musical scales (Lega et al., 2016). This type of higher-order action–perception representation resembles a hierarchical auditory sequence representation that exists in the primary auditory cortex beyond simple pitch class combinations (Schindler et al., 2013). Therefore, we suggest that the altered notes in our study may have violated the implicit action–perception coupling not only at the individual note-to-key level but also at the sequence level as the improvised sequence comes online. Observing FRN during improvisation thus demonstrates evidence of the *instant* action–perception coupling at the sequence level.

The FRN was substantially attenuated while participants heard a deviant pitch in an improvising partner's melody. This is not surprising, since the exact melody that will be played by the partner cannot be known beforehand. However, the FRN and P3a from a partner's improvisation while the participant's task was the score was still present, implying that players might have shared some sense of melodic predictability. In other words, even though the improvised melodies are not exactly known, they are not entirely random, possibly due to the implied harmony of the accompaniment, making the pitch shift of the feedback still somewhat detectable as "out of the musical context." Behavioral data also support this possibility, since the correlation of two self-improvised melodies per trial was significantly higher when the partner was playing the score than when the partner was improvising (Figure 2a). This suggests that when both pianists were improvising together, they were prioritizing their own melodies less and attending more to their partners' phrases. This may also explain the existence of the FRN upon hearing the partner's improvisation while the self-task was to play the score as the melodic correlation between the partner's two improvised melodies was higher, leading to better predictions of the partner's improvised notes. The P3a may also reflect this level of anticipation for improvised melodies. Additionally, P3a typically has a larger amplitude with slower wave morphology, allowing the response to be more easily detected against the EEG noise floor.

The agency effect (i.e., self-other difference) was most reliably found when we examined the response of scored melodies. It is demonstrated by the significantly larger FRN, P3a, and P3b response to altered feedback in self-generated action compared to other-generated action, which replicates previous findings in turn-taking musical tasks (Huberth et al., 2019; Loehr et al., 2015; Maidhof et al., 2010) and nonmusical observational tasks (Fukushima & Hiraki, 2009; Koban et al., 2012). This suggests that pianists in our study allocated more neural resources for stronger predictions, attention, and subsequent memory update processing for self-generated feedback than other-generated feedback. These clear effects of agency in turn-taking piano duet tasks demonstrate a contrast with the lack of agency effect found by Loehr et al. (2013). In their task, the piano duet was designed with simultaneous actions from duet partners, each of whom played two notes comprising a four-note chord together. However, it is unclear whether the temporal structure of actions by itself is the direct cause of this discrepancy in the agency effects. Regardless of whether the action is simultaneous (Novembre et al., 2014) or turn-taking (Hadley et al., 2015), temporal coordination accuracy was disrupted by interference in motor simulation. Nonetheless, the agency effect in our results suggests more salient performance monitoring of their own actions than those of others.

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The FRN data showed a significant interaction between the task condition and task similarity between partners. Given the attenuation of FRN for altered feedback in the other's improvisation, we further investigated how the data in question were affected by the joint task context separately. Specifically, when looking at how the same monitoring process is modulated differently by the concurrent partner's task, two interesting patterns emerge. First, the partner's task type modulated the FRN amplitude of self-generated improvised melodies such that the FRN was larger when the partner was playing the score instead of improvising. Based on our behavioral data (Table 1), the number of unique melodies and notes played by a participant did not differ whether their partner was playing the score or improvising. Since the partner's task type does not affect the quality of produced improvisation, the difference may be more rooted in processing the partner's melodies itself. Second, the player's own task type modulated the FRN amplitude of other-generated score melodies, as the FRN was larger when the player was playing the score instead of improvisation. Thus, like the above scenario, the monitoring process for the other's action outcome may be the source of this difference.

How would the concurrently ongoing improvisation reduce FRN-related neural resources? One possible explanation is that simply hearing more task-relevant novel stimuli by improvisation requires more encoding and memory-updating computations than hearing precomposed melodies, thus taking neural resources from the FRN-related monitoring processes. Attenuation of FRN has been observed when participants were introduced to a higher cognitive load (trial-based feedback on performance and eye movement) than a lower cognitive load (trial-based feedback on performance only) (Krigolson et al., 2012, 2015). In our task, during encoding auditory information of one's own or a partner's newly improvised melodies, performing the joint task and monitoring the outcome may have been burdened by building the global structure of the music in each trial. It is also possible that encoding improvisatory melodies may require increased working memory capacity or attention, which takes away the neural resources available for the early performance monitoring that holds predicted outcomes and compares them to the incoming sounds. Analogous attenuation of MMN has been observed when a distraction is introduced compared to active participation in the task (Schröger et al., 2000), especially when the distraction is task-relevant (Berti, 2008; Horváth & Winkler, 2010; Roeber et al., 2009). However, our behavioral data do not provide direct evidence for these two possible explanations as the IOI differences were similar regardless of partner's role. Future research can differentiate task demands and improvisations while manipulating the level of complexity or structural levels required for improvisation. Nonetheless, our ERP results suggest that musical improvisation affects the overall effectiveness of the early performance monitoring process.

Interestingly, the task similarity between partners affected the later process more straightforwardly, as the P3b response was greater when two players were engaged in the same task (e.g., score-score or improv-improv) than when they were in different tasks (e.g., score-improv or improv-score). This effect was clearly observed regardless of the difference between score and improvisation or the difference between self and other.

In general, the P3b response is considered to reflect temporoparietal attention-related activity and its subsequent memory processing related to task relevance (see Polich, 2007 for review). The larger P3b observed in the same-task duet suggests that, in a joint task where the tasks are similar between partners, the relevance of the partner's action is enhanced for both performers. This contrasts with our previous piano-duet turn-taking study (Huberth et al., 2019), in which no such P3b contrast was observed when partners played scored melodies with similar or dissimilar pitch contours. The discrepancy may have arisen because the two studies vastly differ in terms of the level of similarity between the partners' tasks. Huberth et al. (2019) used scored melodies for both similar and dissimilar tasks, whereas in this study, the difference is between scored and improvised melodies. Thus, our P3b differences may be related to the differences between playing scored pitches and choosing pitches on the fly. If both musicians are playing the known score even if the duet parts are dissimilar, understanding of each other's contribution to the well-defined musical outcome would not be affected. On the other hand, if both are improvising like in our current task, this would be reminiscent of "trading fours" in jazz, where players sequentially take improvisational solos for a short duration as a form of collective creation. If two interleaving melodies do not coherently contribute to the musical outcome, the pianists may feel less musically engaged or think of the musical turn-taking as less successful. Such could easily be the case when one plays the score while the other improvises, with potentially poorly integrated musical outcomes. As P3b is sensitive to shared representations of each other's actions (Demiral et al., 2016; Kato et al., 2016; Peterburs et al., 2019) and joint action planning (Kourtis et al., 2013), we propose that the level of jointness modulates the later stages of performance monitoring in collaborative music-making tasks.

While investigation into improvisational expertise was not the main intention of our study, experienced improvisers and inexperienced improvisers showed a few interesting differences in behavioral and ERP results. First,

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only experienced improvisers showed higher post-deviant slowing and higher poststandard rushing in improvisation than in score, driving the effects observed when all pianists combined. This suggests that altered feedback in improvisation was more salient, and the overall process of improvisation was different from score for experienced improvisers. Second, only inexperienced improvisers showed higher P3a for self-score than self-improvisation. This suggests that the involuntary attention orientation for altered feedback in score was higher for those with score-based training, but this raises the question of why P3a from improvisation was not higher for experienced improvisers. Previous studies that looked into improvisational expertise (Donnay et al., 2014; Goldman et al., 2020; Limb & Braun, 2008; Lopata et al., 2017; Przysinda et al., 2017) recruited more distinctive groups such as professional jazz musicians. As expertise in improvisation was not part of our hypothesis or recruiting criteria, our participants were mostly score-trained pianists with modest experience in improvisation. Thus, our current paradigm can be used for more rigorous group comparisons based on score-playing and improvisational expertise for both behavior and neural correlates.

For overall ERP results, three other negative potentials associated with expectancy violations in the auditory domain may have contributed to our results. First, mismatch negativity (MMN) indices irregularity embedded in an oddball paradigm and functions as a probe to auditory memory (for reviews, see Garrido et al., 2009; Gomes et al., 1995; Näätänen et al., 2007). In our study, repeatedly played score conditions likely enhanced pianists' transient auditory memory, thus additional MMN contribution could explain the higher FRN observed in the score conditions than the FRN in the improvisation condition. However, if MMN is the main contributor to the negativities in all cases rather than the action-related FRN, it fails to explain why we still observed the negativity in the improvisation condition, since almost all improvised melodies were novel and independent of auditory memory trace. Second, early right-anterior negativity (ERAN) reflects expectancy violations in music-syntactic information such as tonal irregularities (Koelsch et al., 2000, 2003). However, it is important to emphasize that our deviant pitches never introduced out-of-key notes. Out-of-key notes were only possible during improvisation but were seldom used and irrelevant for the deviant-related negativity because deviant pitches were always chosen only from in-key notes. Thus, ERAN's contributions are unlikely. Third, N2b reflects controlled registration of mismatch detection on task-relevant targets (Folstein & Van Petten, 2008; Horváth et al., 2008; Korka et al., 2022; for reviews, see Ritter et al., 1992). In our study, despite our task instruction (to ignore the deviant pitches and continue playing to achieve a joint musical outcome), our pianists slowed down after hearing a deviant note, suggesting that deviant detection was not entirely task-irrelevant for them. Thus, N2b may have contributed to our observed negativity. Interestingly, however, our behavioral data showed that post-deviant slowing was more significant for improvisation conditions than score conditions, which suggests that deviant detection was more salient when pianists improvised. If N2b is the main contributor to the negativity, our data should have shown larger negativity in improvisation than score conditions, which is the opposite of what we found. At least we suggest that the N2b contribution to the observed negativity would be relatively small especially for the improvisation condition.

The current study comparing improvisation with score performance has some limitations. First, the score was repeated by the pianists throughout their session while improvisation occurred on the fly for each trial. Clearly, having multiple score conditions may reduce the impact of this problem. On the other hand, if multiple scores were employed, the number of extensive repetitions for observing reliable ERPs and the cognitive load of sight-reading needs would complicate the paradigm even more. Second, the turn-taking was relatively short, as each player had only two phrases to generate material during improvisation. This might have limited pianists' creative idea generation even though all 61 keys of the keyboard were allowed. A longer improvisation could engage musicians in a more natural mode of idea generation, which would allow us to observe how their performance monitoring strategies evolve differently over a course of shared improvisation. In addition, deviant pitches only occurred on fourth and fifth notes of the two phrases, which may result in enhanced expectations of altered pitch (possible attenuation of FRN), especially if the deviant pitch did not occur in the first phrase. Due to the limited number of overall trials within the session, the current data do not allow such comparison while the awareness of the deviant occurrence affecting the performance monitoring process remains an interesting question. Third, our study implemented constrained improvisation where players could only change the pitch. This was to simplify the task for untrained improvisers and to keep the rate of auditory stimulus near constant to prevent confounding effects on ERPs from rhythmic variations. In fact, the overall number of unique melodies and notes generated by trained and untrained improvisers did not differ. Thus, the simplicity of our improvisation task was on the one hand advantageous for all pianists, who successfully exchanged simple melody ideas and jointly evaluated the musical goals regardless of experience and without lengthy practice. This afforded us to answer our fundamental questions about performance

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monitoring during improvisation without employing a highly stylized improvisation context and the expert performers who specialize in that style. On the other hand, improvisation experts may in fact employ a wide range of mental schemata of music that fit various situations and know where to deploy them appropriately. As ERP measurement inherently requires trial repetitions to obtain a sufficient response, substantial challenges exist for making joint improvisation tasks naturalistic, meaningfully creative, and engaging every single time. Fourth, our study did not use eye-tracking or record video footage to measure the potential effects of eye movements and visual processing during performance tasks. In the future, these visual measurements may reveal further behavioral differences between score and improvisation performances, such as number of saccades, timing, and direction. Fifth, our results showed a marginal correlation between the emotional reactivity subscore and smaller FRN amplitude in reaction to self-played melodies (Supplementary Material Section 4), similar to our previous study (Huberth et al., 2019). Although this may seem puzzling given the colloquial understanding of empathy as being related to engagement with others, interpersonal relationships among members in a music ensemble may function in a more complex manner because they share common goals, rather than compete to pursue individual interests (see Supplementary Material Section 4 for a more detailed discussion). Future studies with larger sample sizes can specify details of the interaction between musical contexts and trait empathy to better understand how they are associated with musical goals and outcome relevance.

In conclusion, we conducted a turn-taking piano-duet study to examine how pianists respond differently to altered pitch feedback in either scored or improvised duet parts. Our study showed that the level of prediction during early-stage processes in performance monitoring, as indexed by FRN, was enhanced by familiarization with the prescribed score materials compared to improvisation, especially for the partners' feedback processing. Most importantly, our results demonstrated that familiarity with action sequences is not a prerequisite for action–perception coupling for self-generated action, as performance monitoring on a sequence-level melodic production works similarly during the real-time improvisational performance. In addition, during a turn-taking musical joint task, more neural resources associated with performance monitoring are allocated to process feedback on one's own performance than feedback on a partner's performance. Moreover, the later-stage attention and memory processes, indexed by P3a and P3b, are increasingly sensitive to the structure of shared tasks seen similarly for joint-improvisation and joint-score playing. Our data also showed that cognitive loads may be increased for both listening to and creating novel melodies that require new encodings, interfering with the available resources for performance monitoring. Lastly, the current results aligned with the finding of our previous study by showing that the higher trait empathy score, especially emotional reactivity subscale, was related to smaller FRN amplitude in response to the feedback to self-generated action. Altogether, the present study contributes to our understanding of neural correlates of musical collaborations by extending the musical context to improvisation.

AUTHOR CONTRIBUTIONS

Kunwoo Kim: Conceptualization; data curation; formal analysis; investigation; methodology; validation; visualization; writing – original draft; writing – review and editing. **Noah Fram:** Conceptualization; data curation; formal analysis; investigation; methodology; validation; visualization; writing – review and editing. **Barbara Nerness:** Conceptualization; data curation; methodology; writing – review and editing. **Cara Turnbull:** Conceptualization; data curation; formal analysis; methodology; writing – review and editing. **Aditya Chander:** Conceptualization; data curation; formal analysis; methodology; writing – review and editing. **Elena Georgieva:** Conceptualization; data curation; methodology; writing – review and editing. **Sebastian James:** Conceptualization; data curation; writing – review and editing. **Matt Wright:** Resources; software; methodology; validation; writing – review and editing. **Takako Fujioka:** Conceptualization; data curation; formal analysis; funding acquisition; investigation; methodology; project administration; resources; supervision; validation; visualization; writing – original draft; writing – review and editing.

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CONFLICT OF INTEREST STATEMENT

The authors declare no conflicts of interest.

DATA AVAILABILITY STATEMENT

The data that support the findings of this study are available from the corresponding author upon reasonable request.

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SUPPORTING INFORMATION

Additional supporting information can be found online in the Supporting Information section at the end of this article. **Data S1:** Supplementary Materials.

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